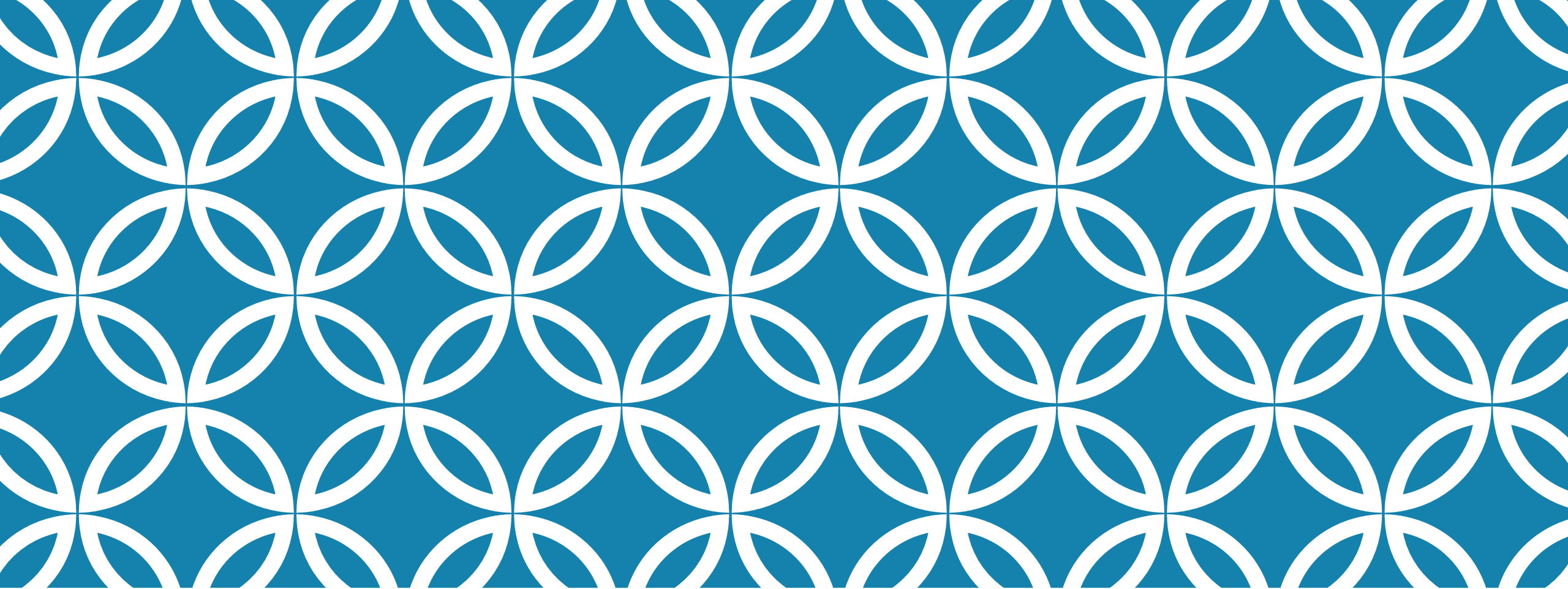




INTRODUCTION TO DATA SCIENCE

OUTLINE

Introduction to Python for Data Science

Discovering the Match between Data Science and Python,

Introducing Python's Capabilities and Wonders, Setting Up Python for Data Science

Understanding the Tools, Working with Real Data, Conditioning Your Data, Shaping Data, Putting What You Know in Action

Getting a Crash Course in Matplotlib, Visualizing the Data

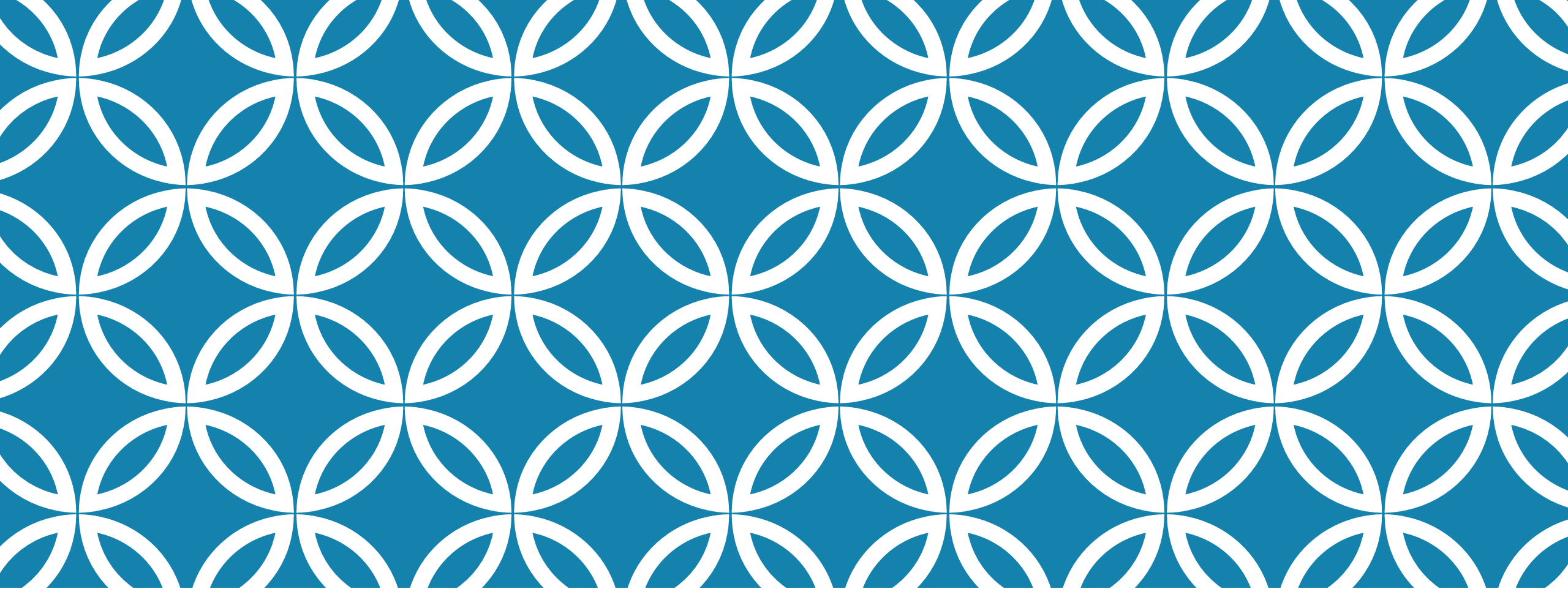
Stretching Python's Capabilities, Exploring Data Analysis, Reducing Dimensionality, Clustering, Detecting Outliers in Data

Ten Essential Data Resources, Ten Data Challenges You Should Take

VIDEO LINK

<https://www.youtube.com/watch?v=jNeUBWrrRsQ&t=146s>

<https://www.youtube.com/watch?v=mkv5mxYu0Wk&t=253s>



DATA SCIENCE

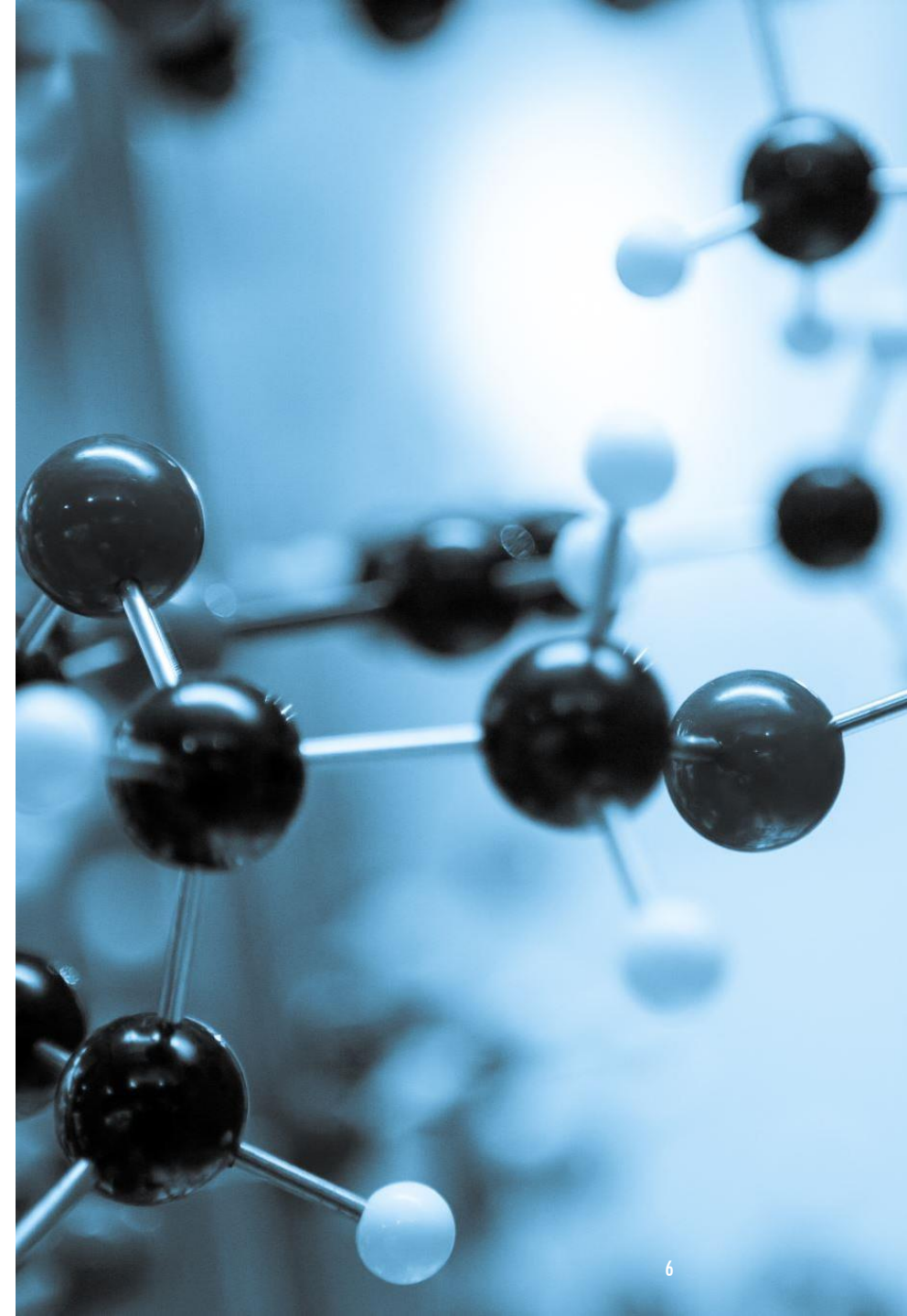
An Overview

INTRODUCTION TO DATA SCIENCE

Data Science is a multidisciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data.

IMPORTANCE OF DATA SCIENCE

Data Science is important because it helps organizations make better decisions, improve operational efficiency, identify new opportunities, and enhance customer experiences.



HISTORY OF DATA SCIENCE

The history of data science is quite fascinating and spans several decades. Here's a brief overview:

Early Beginnings

1960s: The term "data science" was first coined in the early 1960s to describe a new profession that would support the understanding and interpretation of large amounts of data¹

John Tukey's influential paper, "The Future of Data Analysis," highlighted the merging of statistics and computers¹

Evolution and Growth

1980s-1990s: With the advent of computers, data science began to evolve rapidly. The development of databases and data warehousing technologies allowed for more efficient data storage and retrieval²

2000s: The rise of the internet and the explosion of data led to the emergence of big data. Companies started to realize the potential of data-driven decision-making

Modern Data Science

2010s-Present: Data science has become a multidisciplinary field, incorporating elements of statistics, computer science, and domain-specific knowledge²

The development of machine learning and artificial intelligence has further expanded the capabilities of data science

A Brief History of Data Science



Data Scientists

expanded by 15,000% in between 2011 – 2012.

2011

Title "Data Scientist"

turned into a trendy expression and in the long run a piece of the language.

2008

Hadoop 0.1.0,
an open-source and non-relational
database was released

2006

International Council for Science:

Committee on Data for Science
and Technology started
distributing the **Data Science
Journal**

2001

Jacob Zahavi brought
up the requirement for new devices to
deal with the enormous measures of
data accessible to organizations.

1999

Gregory Piatetsky – Shapiro

arranged a Knowledge Discovery in
Databases workshop.

1994

Business Week ran
the main story,
Database Marketing

1989

Peter Naur

wrote the Concise Survey of
Computer Methods, utilizing the
expression "Data Science,"
over and over.

1977

John Wilder Tukey

composed a second paper titled
"Exploratory Data Analysis"

1974

1962

John Wilder Tukey

expounded on a move in the world of
statistics in 1962.



DatabaseTown.com

History of data science

Corporate,
Industry &
Business
People



1955s

Taylorism was adopted majorly across industrial revolution



1965s

Peter Drucker introduced the idea - knowledge worker



1975s

Kaizen and Toyota Production System is being adopted in manufacturing



1985s

Excel became the go to tool among knowledge workers

Academia,
Engg,
Math,
Computer
Hobbyists



1955s

Computers and AI were extensively researched after World War 2



1965s

The HP 9100A, the world's first desktop computer



1975s

John Tukey through his book 'Exploratory Data Analysis' showed the importance of visualizing data



1985s

Operating Systems made its debut in the market with MS Windows

1995s

Internet makes the world connected

2000s

DOTCom Boom - every company take their business online

2005s

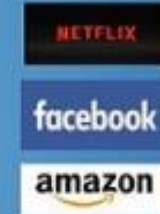
Service industry transformed customer experience and they improved with data

2010s

Smartphones intensified customer obsession and era of on-demand economy starts

2015s

Machine Learning and AI becomes mainstream. Computation intensified with GPUs



KEY MILESTONES

Big Data: The exponential growth of data volumes has driven the need for advanced data processing and analysis techniques¹

Machine Learning: The integration of machine learning algorithms has enabled more accurate predictions and insights from data

AI and IoT: The advent of artificial intelligence and the Internet of Things (IoT) has further expanded the applications of data science

Data science continues to evolve, playing a crucial role in various industries, from healthcare to finance, and driving innovation and efficiency in numerous fields.

DATA SCIENCE PROCESS



The Data Science process involves several steps:



1. Data Collection



2. Data Cleaning



3. Data Exploration and Analysis



4. Data Modeling



5. Data Visualization



6. Deployment and Maintenance

DATA SCIENCE COMPONENTS

1. Data Collection and Management

- **Description:** Gathering raw data from various sources such as databases, APIs, sensors, or web scraping, and storing it securely for analysis.
- **Example:** Collecting transaction records from an e-commerce platform





2. Data Preparation and Cleaning

- **Description:** Cleaning and preprocessing data by handling missing values, removing duplicates, and standardizing formats to ensure accuracy and consistency.
- **Example:** Removing incomplete entries from a dataset of customer feedback



3. Data Analysis

- **Description:** Applying statistical and mathematical methods to explore the data, identify patterns, and draw conclusions.
- **Example:** Using regression analysis to determine factors influencing sales performance

4. Machine Learning and Modeling

- **Description:** Building predictive or descriptive models using machine learning algorithms to solve specific problems.
- **Example:** Creating a recommendation engine for an online streaming service





5. Data Visualization

- **Description:** Presenting data insights through charts, graphs, and dashboards to make findings comprehensible and actionable.
- **Example:** Visualizing website traffic trends using a line chart in Tableau

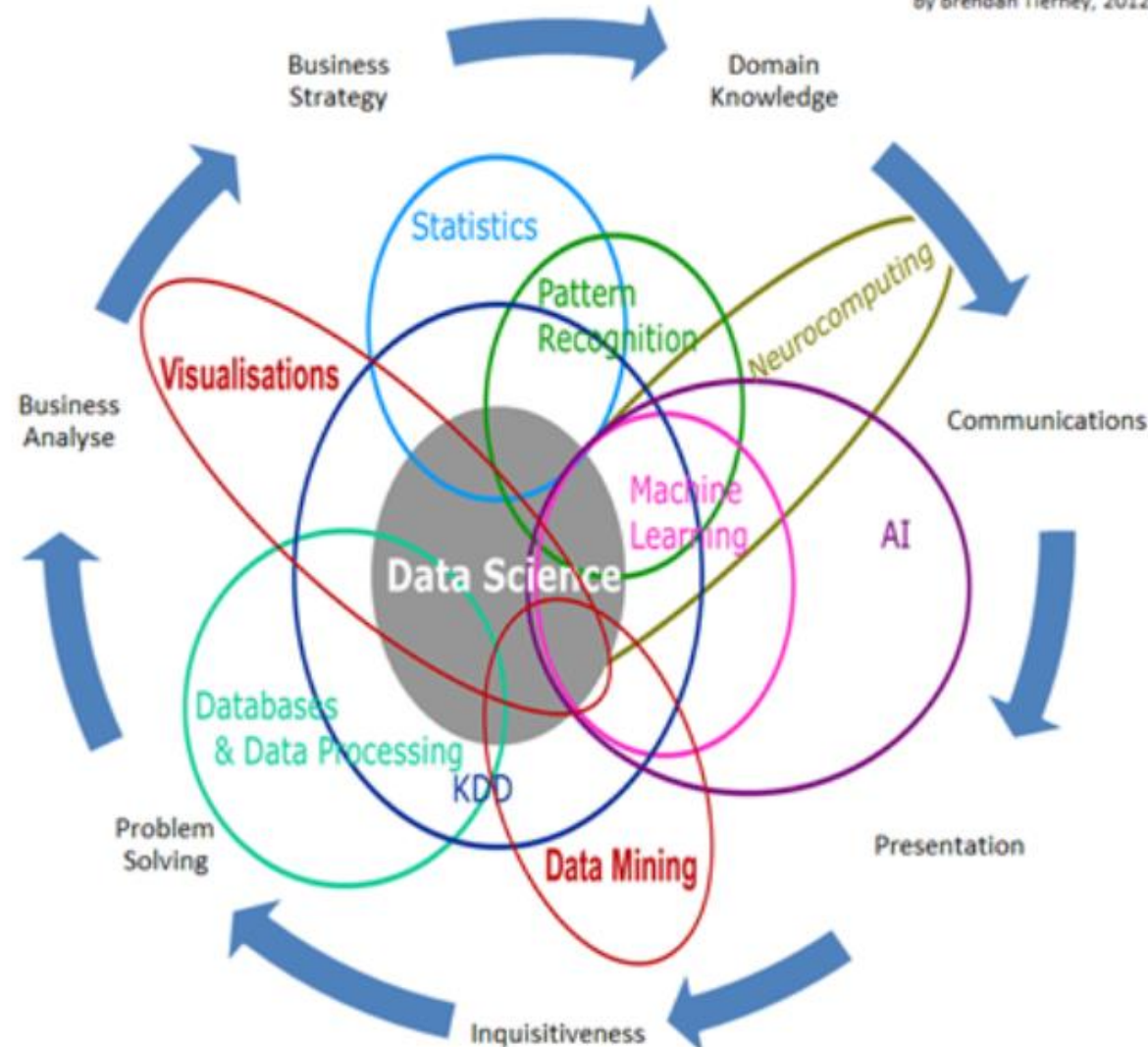


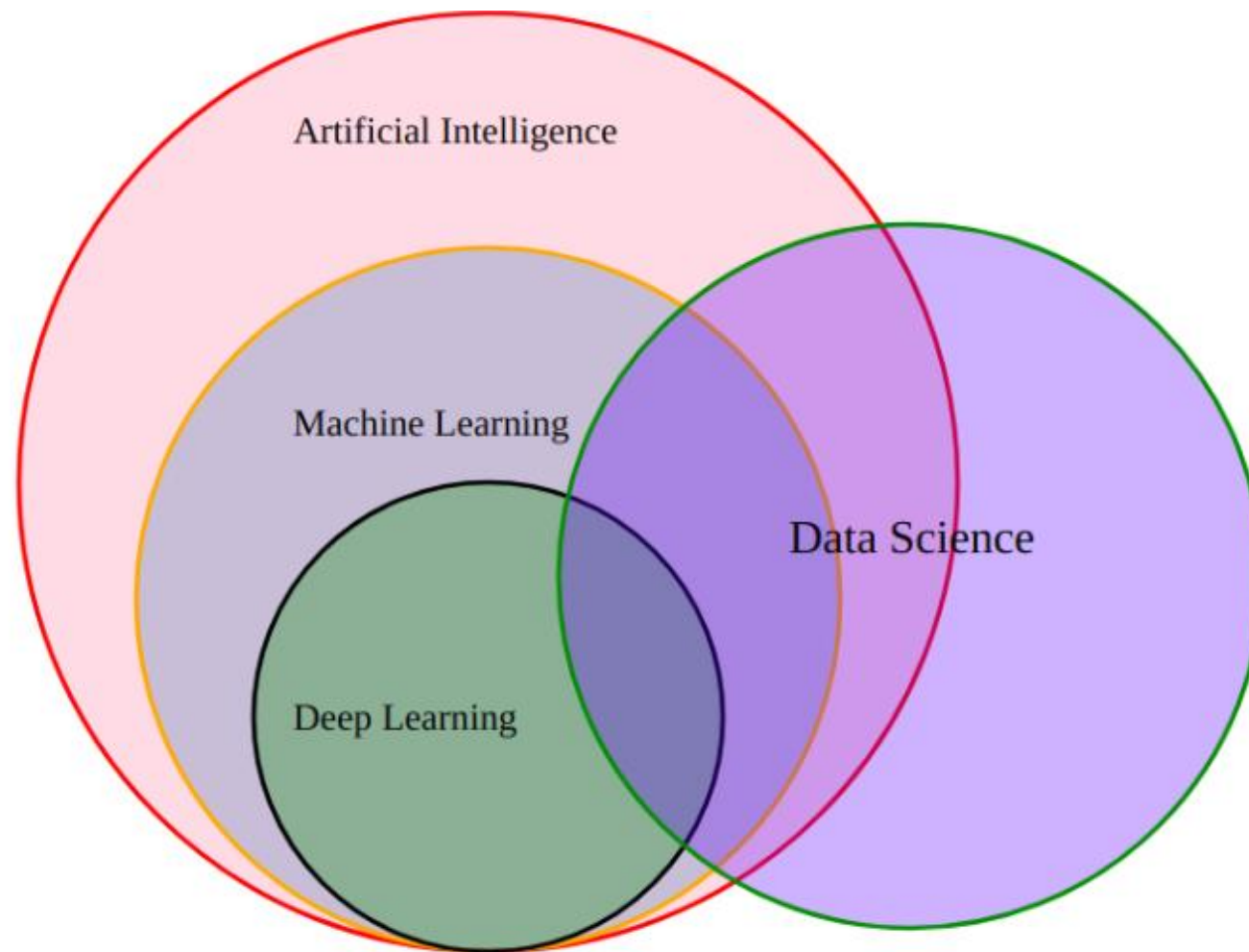
6. Communication and Decision-Making

- **Description:** Translating analytical findings into insights and recommendations for stakeholders or decision-makers.
- **Example:** Presenting the impact of a new pricing strategy to a company's executive team

Data Science Is Multidisciplinary

By Brendan Tierney, 2012





<https://gist.github.com/krajiv26/839ea54316a257c1fef797deb4d13a8b>

TOOLS AND TECHNOLOGIES

Some popular tools and technologies used in Data Science include:

- Programming Languages: Python, R
- Data Manipulation: Pandas, NumPy
- Data Visualization: Matplotlib, Seaborn
- Machine Learning: Scikit-learn, TensorFlow, Keras
- Big Data: Hadoop, Spark

APPLICATIONS OF DATA SCIENCE



Data Science has a wide range of applications in various industries, including:



- Healthcare: Disease prediction, personalized medicine



- Finance: Fraud detection, algorithmic trading



- Marketing: Customer segmentation, sentiment analysis



- Retail: Inventory management, recommendation systems



- Transportation: Route optimization, autonomous vehicles

APPLICATIONS

Data science has a wide range of applications across various industries. Here are some notable examples:

1. Healthcare

- **Disease Prediction and Diagnosis:** Data science helps in predicting diseases and diagnosing conditions at an early stage using patient data and medical records¹
- **Personalized Medicine:** Algorithms analyze patient data to recommend personalized treatment plans



2. Finance

- **Fraud Detection:** Data science techniques are used to detect fraudulent activities by analyzing transaction patterns
- **Risk Management:** Financial institutions use data science to assess and manage risks

3. Marketing

- **Customer Segmentation:** Data science helps in segmenting customers based on their behavior and preferences for targeted marketing
- **Sentiment Analysis:** Analyzing customer feedback and social media posts to gauge public sentiment towards products or brands



4. Retail

- **Inventory Management:** Data science optimizes inventory levels by predicting demand and managing stock
- **Recommendation Systems:** E-commerce platforms use data science to recommend products to customers based on their browsing and purchase history



5. Transportation

- **Route Optimization:** Data science is used to optimize shipping routes and reduce delivery times
- **Autonomous Vehicles:** Self-driving cars rely on data science for navigation and decision-making



6. Sports

- **Performance Analysis:** Data science evaluates athletes' performance and helps in strategizing games
- **Injury Prevention:** Analyzing data to predict and prevent injuries in athletes

7. Government

- **Tax Evasion Prevention:** Data science helps in identifying and preventing tax evasion
- **Public Safety:** Predictive policing and crime analysis are enhanced using data science



8. Social Media

- **Content Recommendation:** Platforms like Facebook and Instagram use data science to recommend content to users
- **User Behavior Analysis:** Analyzing user interactions to improve user experience and engagement

9. Gaming

- **Game Development:** Data science is used to analyze player behavior and improve game design
- **Cheat Detection:** Identifying and preventing cheating in online games



10. Education

- **Personalized Learning:** Data science helps in creating personalized learning experiences for students
- **Performance Tracking:** Analyzing student performance data to identify areas of improvement

These applications demonstrate the versatility and impact of data science in solving real-world problems and driving innovation across various sectors.

WHY IS PYTHON SUITED FOR DATA SCIENCE?

Article read:

<https://learnpython.com/blog/why-python-used-for-science/>

DATA SCIENCE IN PYTHON



TensorFlow



matplotlib



SciPy

PySpark



Keras



PyTorch



NumPy



scikit
learn

PYTHON MARKET SHARE

<https://cdn-cjmik.nitrocdn.com/UjszoEMIGzQLBmRYICliaPmdTnvQlovN/assets/images/optimized/rev-b7b1dec/www.aalpha.net/wp-content/uploads/2019/10/python-market-share-2019.gif>

CHALLENGES IN DATA SCIENCE

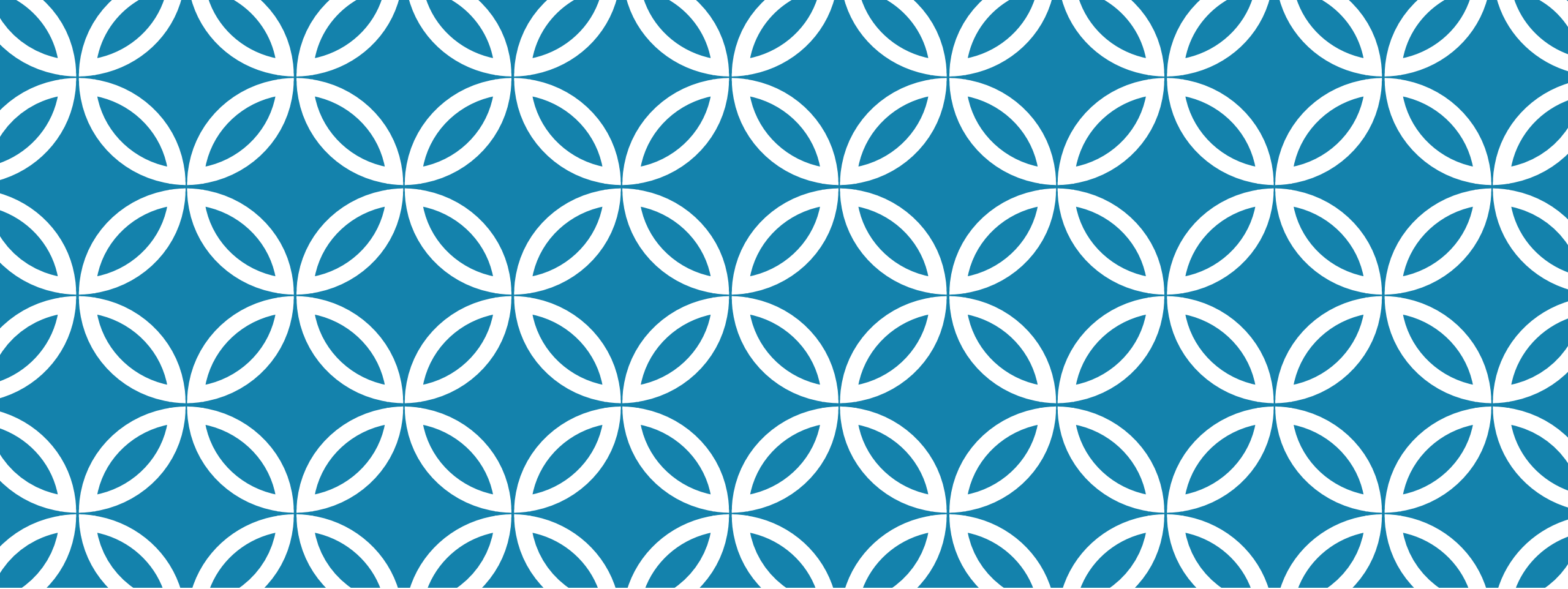
Some common challenges faced in Data Science include:

- Handling large volumes of data
- Ensuring data quality and accuracy
- Dealing with unstructured data
- Addressing privacy and security concerns
- Keeping up with rapidly evolving technologies and methodologies



FUTURE OF DATA SCIENCE

The future of Data Science looks promising with advancements in artificial intelligence, machine learning, and big data technologies. It will continue to play a crucial role in driving innovation and solving complex problems across various domains.



BIG DATA

An Overview

INTRODUCTION TO BIG DATA

Big Data refers to the large volume of data that is generated, stored, and analyzed by organizations to gain insights and make better decisions. It encompasses structured, semi-structured, and unstructured data.

CHARACTERISTICS OF BIG DATA



Big Data is often described by the following characteristics:



1. Volume: The vast amount of data generated every second.



2. Velocity: The speed at which new data is generated and processed.



3. Variety: The different types of data (structured, semi-structured, unstructured).



4. Veracity: The quality and accuracy of the data.



5. Value: The insights and benefits derived from analyzing the data.

BIG DATA TECHNOLOGIES

Several technologies are used to handle and analyze Big Data, including:

- Hadoop: An open-source framework for distributed storage and processing of large datasets.
- Spark: A fast and general-purpose cluster computing system for big data processing.
- NoSQL Databases: Databases like MongoDB, Cassandra, and HBase designed for handling unstructured data.
- Data Lakes: Storage repositories that hold vast amounts of raw data in its native format.

BIG DATA ANALYTICS

Big Data Analytics involves examining large datasets to uncover hidden patterns, correlations, and insights. It includes techniques such as:

- Descriptive Analytics: Summarizing historical data to understand what has happened.
- Predictive Analytics: Using statistical models and machine learning to predict future outcomes.
- Prescriptive Analytics: Recommending actions based on predictive insights.

APPLICATIONS OF BIG DATA

Big Data has a wide range of applications across various industries, including:

- Healthcare: Predicting disease outbreaks, personalized medicine.
- Finance: Fraud detection, risk management.
- Marketing: Customer segmentation, sentiment analysis.
- Retail: Inventory management, recommendation systems.
- Transportation: Route optimization, autonomous vehicles.

CHALLENGES IN BIG DATA

Some common challenges faced in Big Data include:

- Data Privacy and Security: Ensuring the protection of sensitive information.
- Data Quality: Maintaining accuracy and consistency of data.
- Scalability: Handling the growing volume of data efficiently.
- Integration: Combining data from various sources for analysis.

FUTURE OF BIG DATA

The future of Big Data looks promising with advancements in artificial intelligence, machine learning, and cloud computing. It will continue to play a crucial role in driving innovation and solving complex problems across various domains.

CASE STUDY

<https://towardsdatascience.com/netflix-recommender-system-a-big-data-case-study-19cfa6d56ff5>